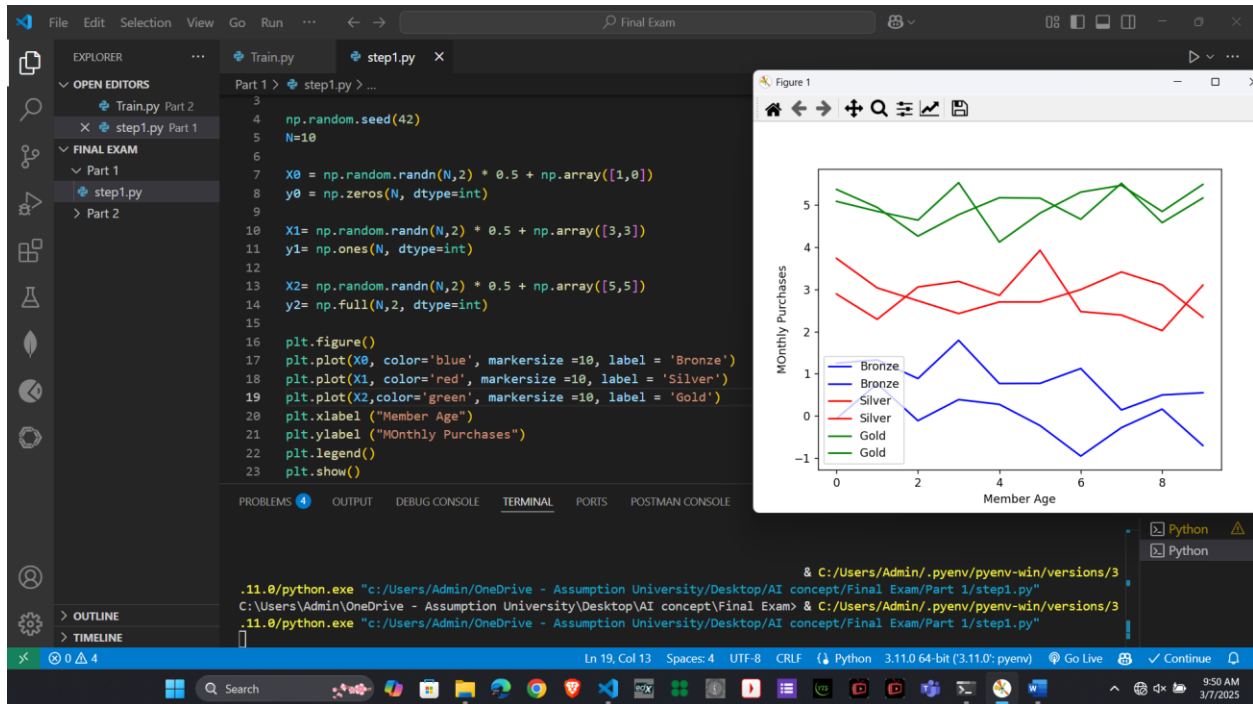
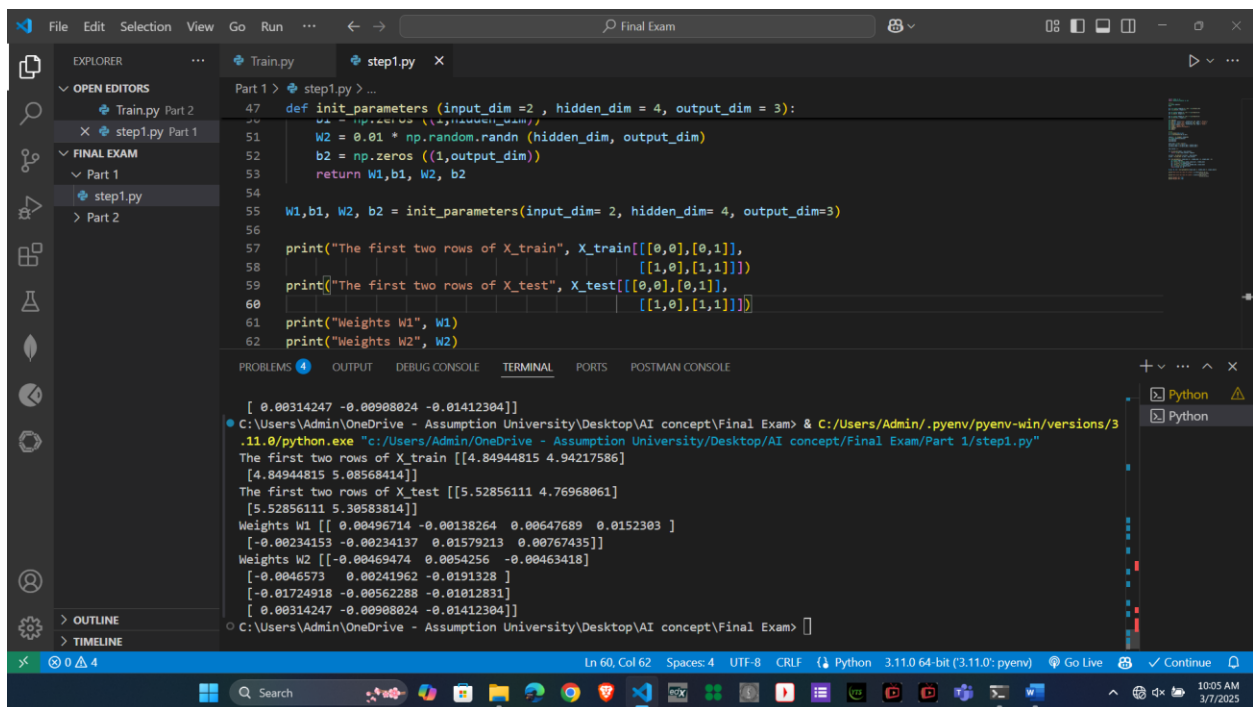


Part 1:

Step 1:



Step2:



Step 3:

The screenshot shows the VS Code editor with the file explorer on the left. The 'EXPLORER' pane shows a project structure with 'Train.py' and 'step1.py'. The 'step1.py' file is open in the editor, showing the implementation of a neural network forward pass and cross-entropy loss function. The code is as follows:

```
Part 1 > step1.py > cross_entropy_loss
67
68 def relu_deriv(z):
69     return (z>0).astype(float)
70
71 def softmax(z):
72     shifted = z - np.max(z, axis = 1, keepdims = True)
73     exp_vals = np.exp(shifted)
74     return exp_vals / np.sum(exp_vals, axis=1, keepdims = True)
75
76 def forward_pass (X, W1, b1, W2, b2):
77     Z1 = X.dot(W1) +b1
78     A1 = relu(Z1)
79     Z2 = A1.dot(W2) + b2
80     probs = softmax (Z2)
81     return Z1, A1, Z2, probs
82
83 def cross_entropy_loss(probs, targets):
84     eps = 1e-12
85     return np.mean(np.sum(targets * np.log(probs + eps), axis= 1))
86
87 Z1, A1,Z2,probs = forward_pass (X (variable) y_train: ndarray[Any, dtype[float64]]
88 loss = cross_entropy_loss(probs, y_train[:2])
```

The terminal pane at the bottom shows the output of the code execution:

```
Z2 value: [[-0.00158159 -0.00156043 -0.00274966]
[-0.00167488 -0.00166398 -0.0029402 ]]
probs value: [[0.33346074 0.3334678 0.33307146]
[0.33347268 0.33347632 0.333051 ]]
Loss: -1.0994289284318544
```

The screenshot shows the VS Code editor with the file explorer on the left. The 'step1.py' file is open in the editor, showing the completion of the neural network implementation. The code is as follows:

```
Part 1 > step1.py > ...
83 def cross_entropy_loss(probs, targets):
84     eps = 1e-12
85     return np.mean(np.sum(targets * np.log(probs + eps), axis= 1))
86
87 Z1, A1,Z2,probs = forward_pass (X_train[:2], W1, b1, W2,b2)
88 loss = cross_entropy_loss(probs, y_train[:2])
89
90 print("Z1 value: ", Z1)
91 print("A1 value: ", A1)
92 print("Z2 value: ", Z2)
93 print("probs value: ", probs)
94 print("Loss: ", loss)
```

The terminal pane at the bottom shows the output of the code execution:

```
[[-0.0046573 0.00241962 -0.0191328 ]
[-0.01724918 -0.00562288 -0.01012831]
[ 0.00314247 -0.00908024 -0.01412304]]
Z1 value: [[ 0.01319334 -0.01818762 0.10859301 0.11248716]
[ 0.01476144 -0.0193312 0.11508968 0.12080433]]
A1 value: [[0.01319334 0. 0.10859301 0.11248716]
[0.01476144 0. 0.11508968 0.12080433]]
Z2 value: [[-0.00158159 -0.00156043 -0.00274966]
Z2 value: [[-0.00158159 -0.00156043 -0.00274966]
Z2 value: [[-0.00158159 -0.00156043 -0.00274966]
Z2 value: [[-0.00158159 -0.00156043 -0.00274966]
Z2 value: [[-0.00158159 -0.00156043 -0.00274966]
[-0.00167488 -0.00166398 -0.0029402 ]]
probs value: [[0.33346074 0.3334678 0.33307146]
[0.33347268 0.33347632 0.333051 ]]
Loss: -1.0994289284318544
```